CFD PROJECT:

TRANSLATING MOVING MESH

CFD MODELLING OF THE DRAFTING EFFECT IN NASCAR



Group 4:

Anthony Laï-Hong-Ting

Morgan Soupama

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INTRODUCTION

→ Project studying problems that involve a translating moving mesh

→ Simulation of the drafting effect between 2 NASCAR cars

I. PROBLEM STATEMENT AND OBJECTIVES

Objectives:

→ Demonstrate that STAR-CCM+ can accurately simulate the drafting effect during a NASCAR overtake using a translating moving mesh.

→ Analyse how the distance between vehicles affects their aerodynamic coefficients (drag, lift)

II. LITERATURE REVIEW AND THEORETICAL DATA

→ Project inspired by the study carried out at Universiti Tenaga Nasional (Malaysia) (https://www.researchgate.net/publication/343391137_CFD_Analysis_of_Slipstreaming_and_Side_Drafting_Techniques_Concerning_Aerodynamic_Drag_in_NASCAR_Racing)

→ Straight-line speed: 89 m/s (322 km/h)

→ Drag coefficient: Cd ≈ 0.45 to 0.5

(https://www.reddit.com/r/NASCAR/comments/1bjjsdd/anyone_know_what_the_drag_coefficient_of_the/)

→ Lift coefficient: -0.46 (downforce) (https://occamsracers.com/2019/09/29/race-cars-drag-and-lift-examples/)

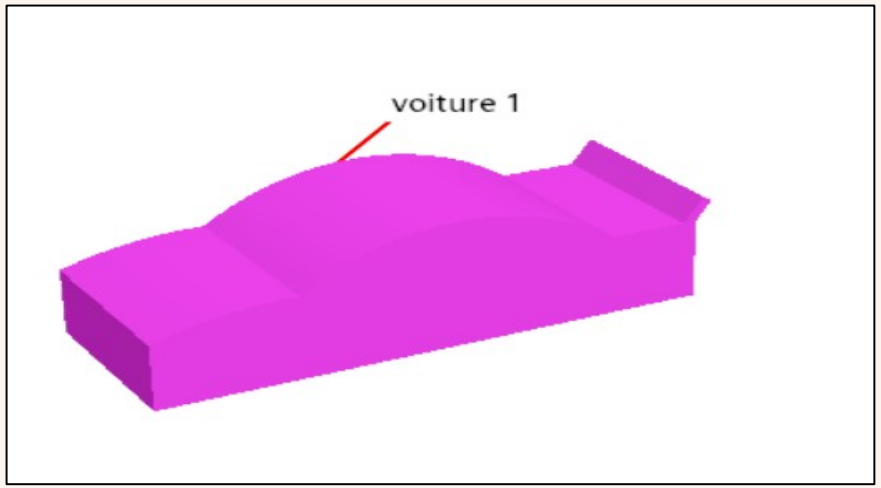
III. STATIC MODELLING IN 2D/3D

1. Model geometry / mesh / physics

→ Dimensions of the simplified car model: 5 x 1.95 x 1.42 m

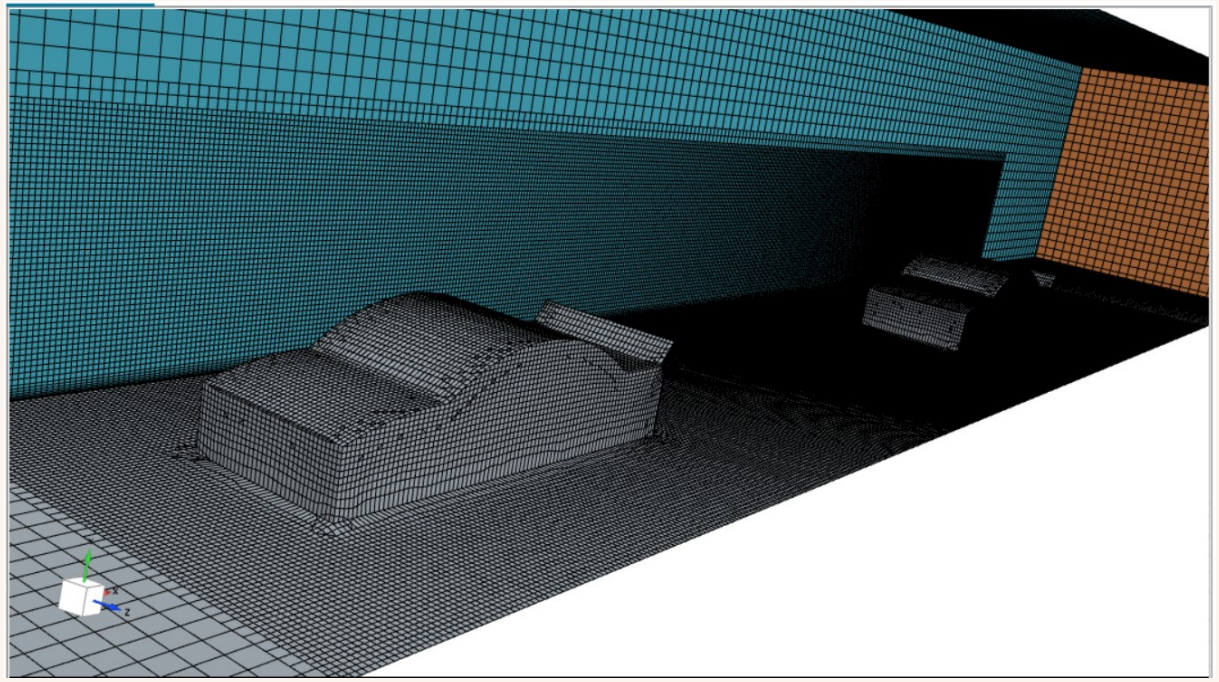
→ Rear wing dimensions: 1.95 x 0.34 m at an angle of 52.69 degrees to the horizontal





III. STATIC MODELLING IN 2D/3D

1. Model geometry / mesh / physics



Mesh parameters

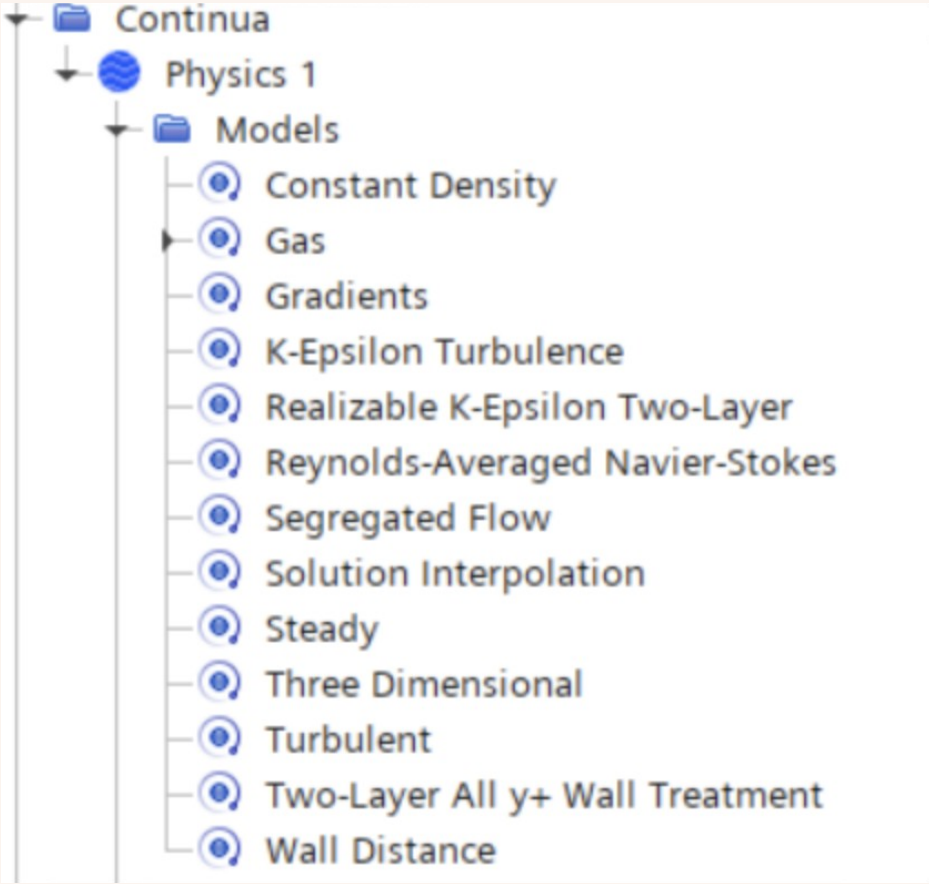
- Base size: 0.5 m
- Number of prism layers: 8
- Prism layer stretching: 1.2
- Total prism layer thickness: 33% of the base size

Simplification:

- Domain reduced to the width of the cars

III. STATIC MODELLING IN 2D/3D

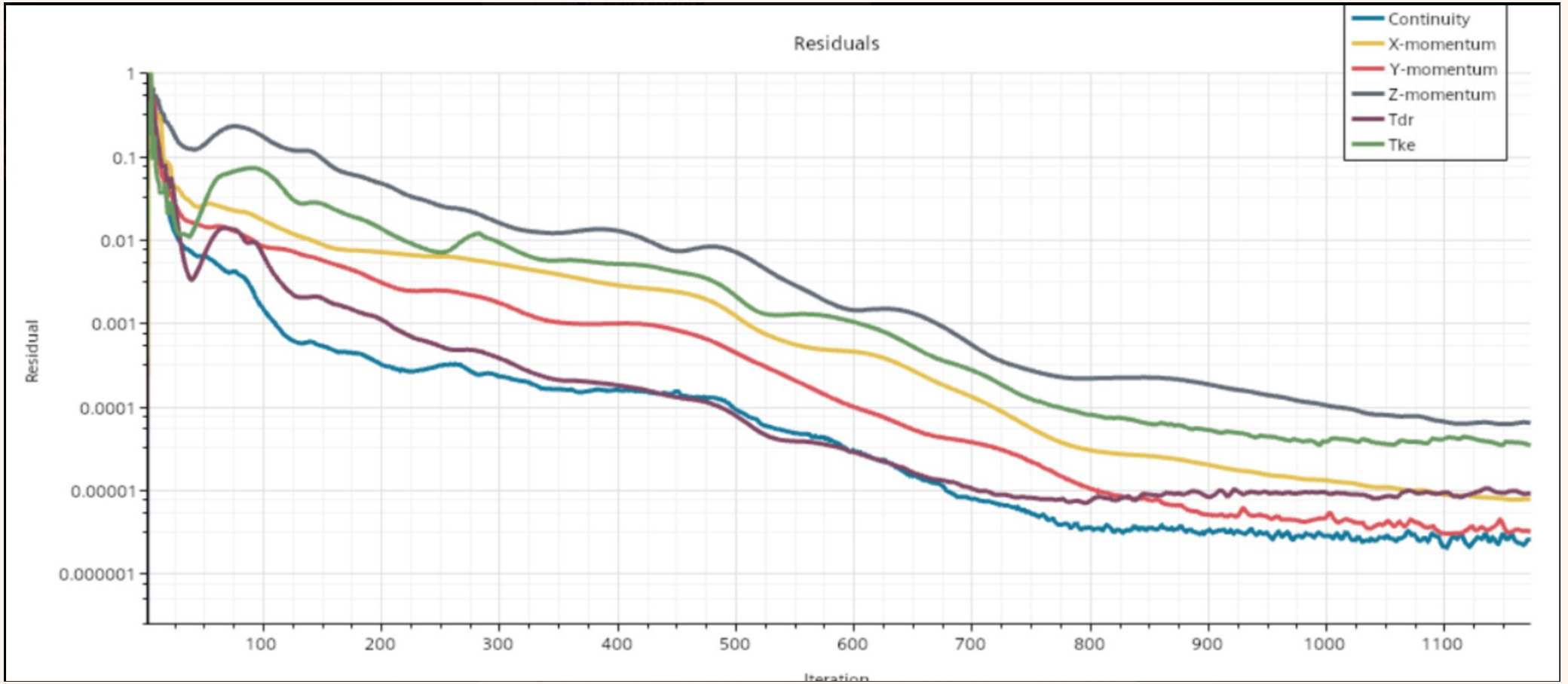
1. Model geometry / mesh / physics



- *Constant-density gas (incompressible air)*
- *3D/2D steady flow*
- *Turbulence modelled with the Realizable two-layer k-epsilon model*
- *"All y+" wall treatment for every y+ range*
- *RANS (Reynolds-Averaged Navier-Stokes) formulation*
- *Segregated solvers for numerical robustness*
- *Overset mesh and local gradients handled through solution interpolation and wall distance*

III. STATIC MODELLING IN 2D/3D
2. 2D simulation - qualitative results

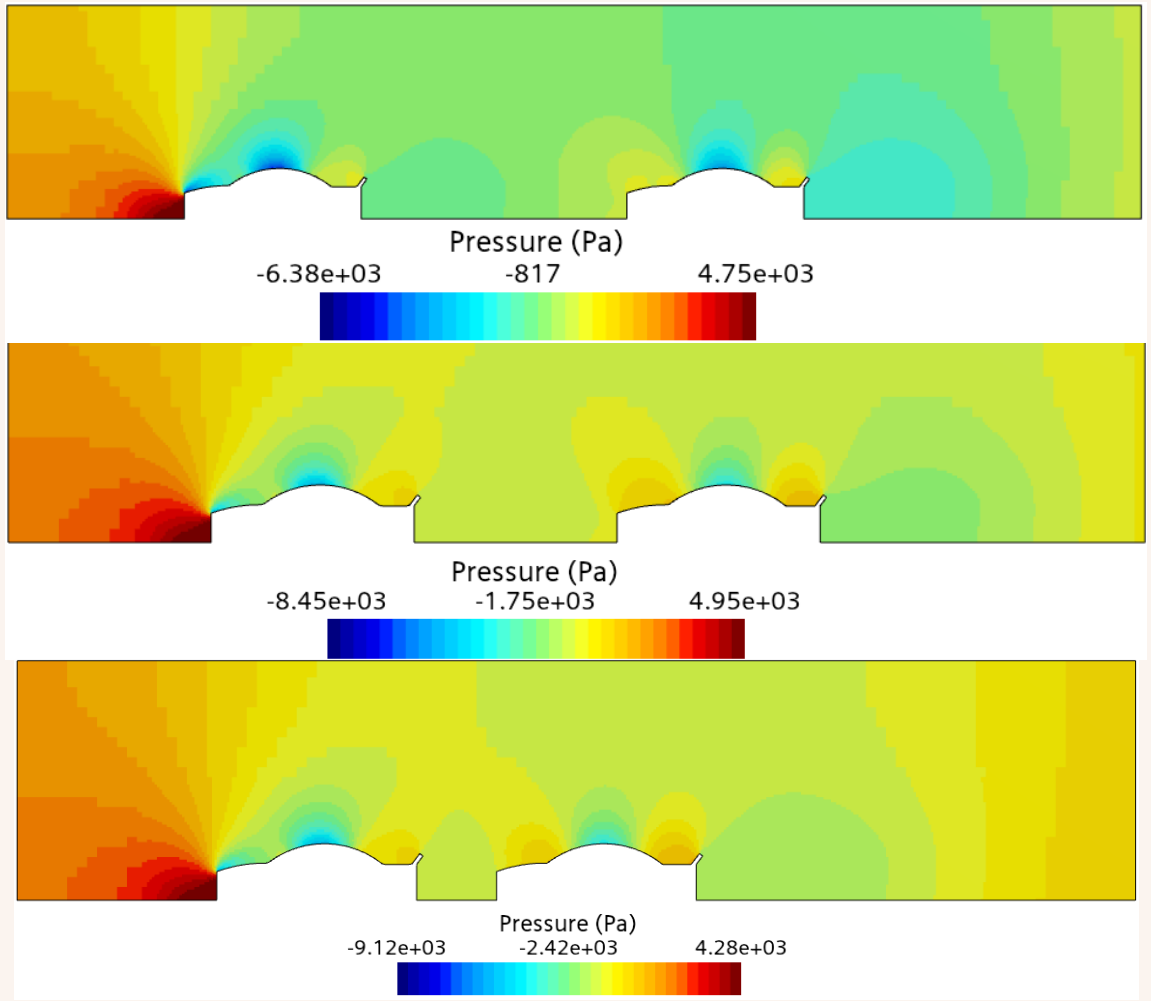
- Residual convergence - static 3D model



III. STATIC MODELLING IN 2D/3D

2. 2D simulation - qualitative results

- Static 2D/3D comparison



7 m gap

5 m gap

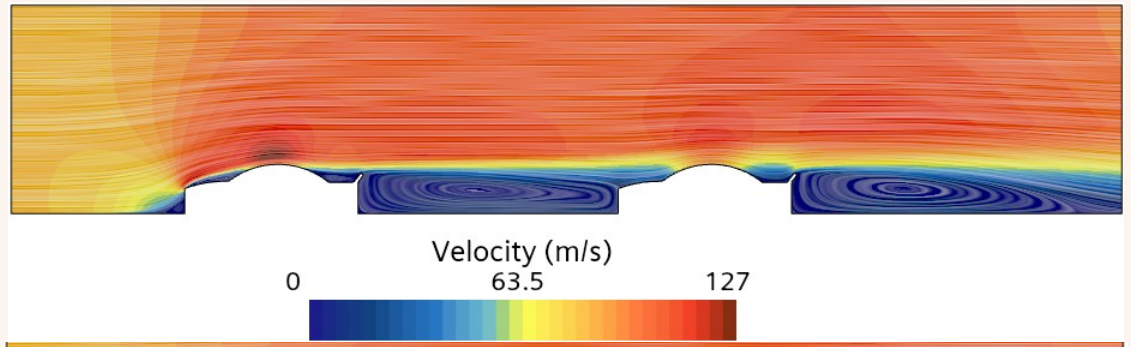
2 m gap

Pressure distribution (Pa) at different positions of the following car

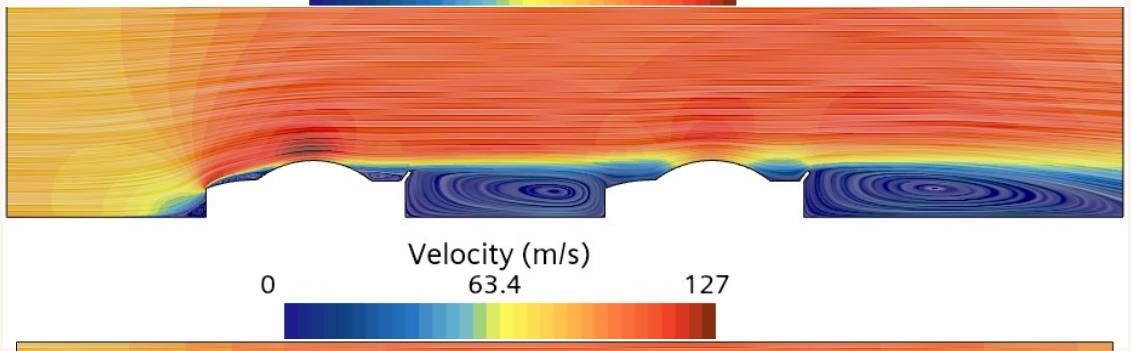
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2. 2D simulation - qualitative results

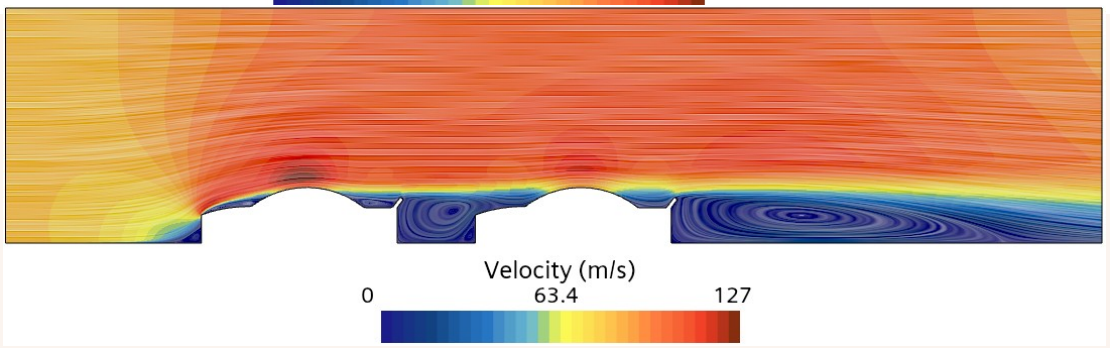
- Static 2D/3D comparison



7 m gap



5 m gap

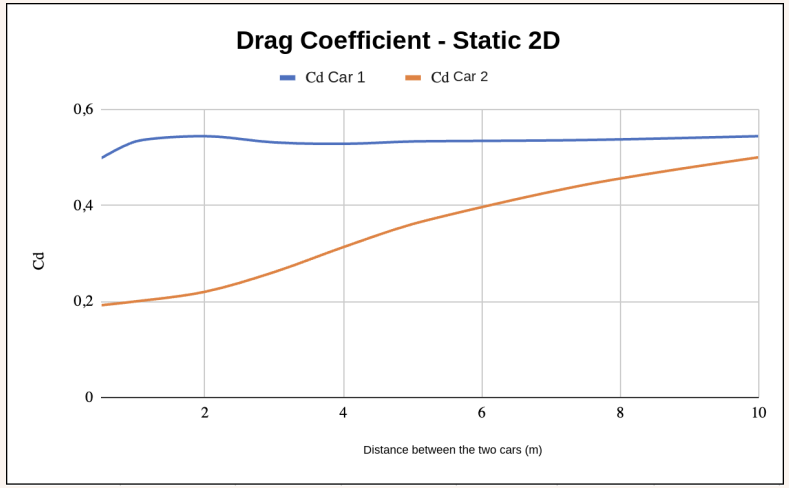


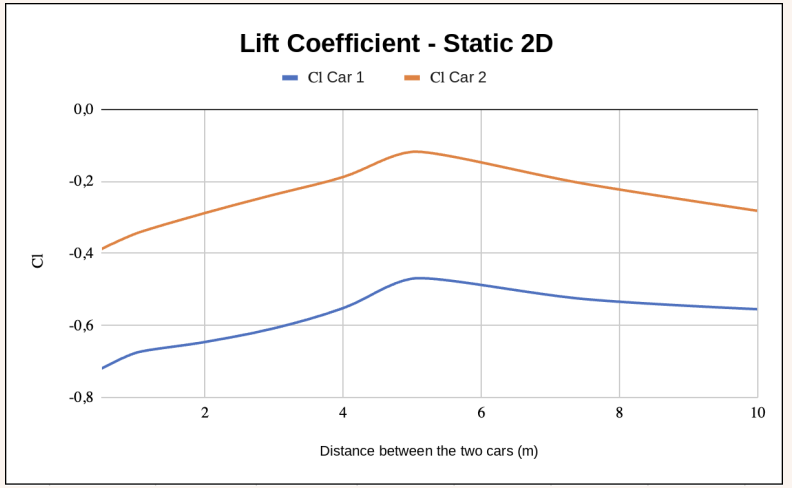
2 m gap

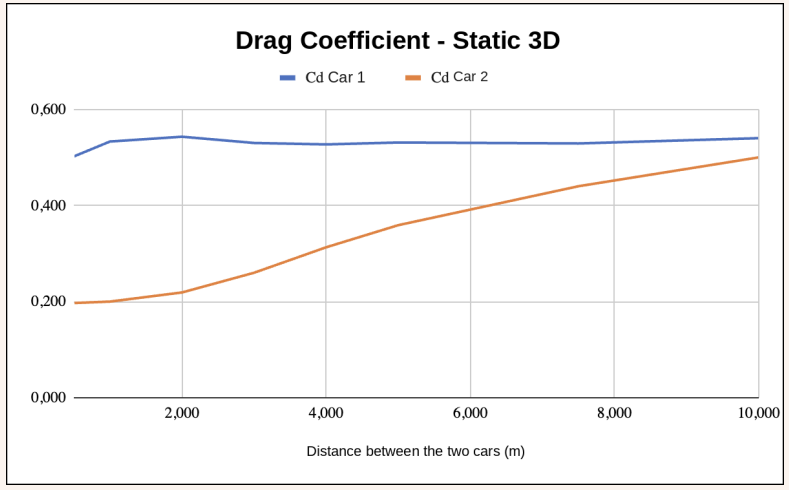
Velocity distribution (m/s) at different positions of the following car

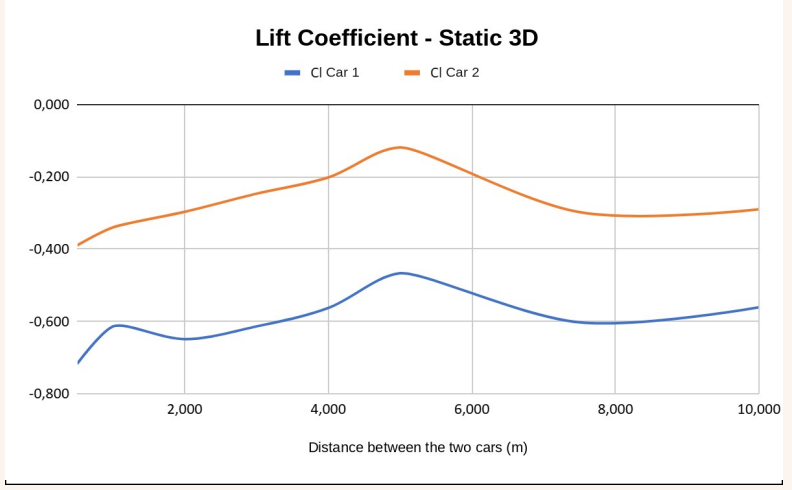
III. STATIC MODELLING IN 2D/3D

2. 2D simulation - quantitative results









IV. DYNAMIC MODELLING IN 2D/3D

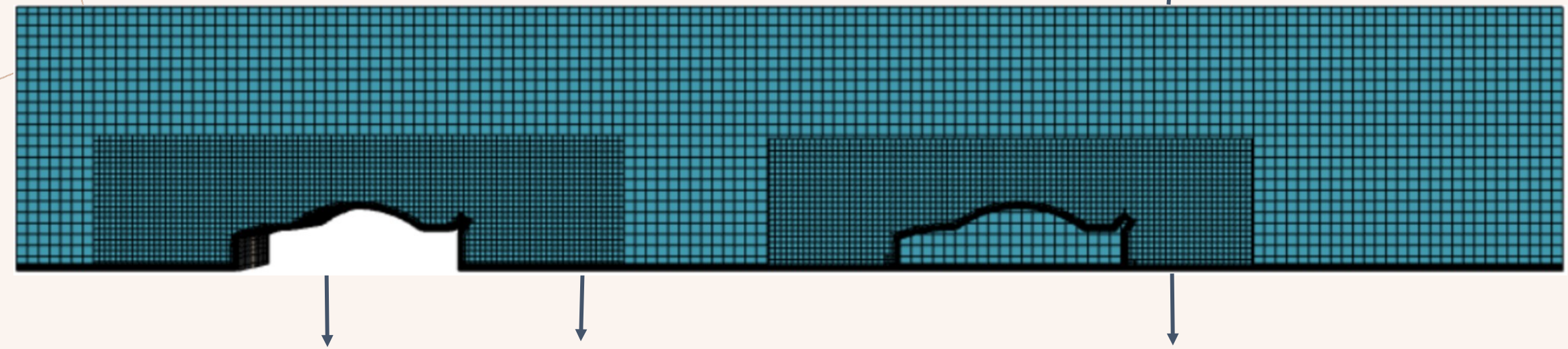
1. Model geometry / mesh / physics

Computational domain

- Base size: 0.25 m
- Number of prism layers: 8
- Prism layer stretching: 1.3
- Prism layer thickness: 0.16 m

Domain

Dimensions:
35 m x 6 m x 1.95 m



Lead car

Custom control

- Base size: 0.15 m
- Dimensions: 12 m x 3 m x 1.95 m

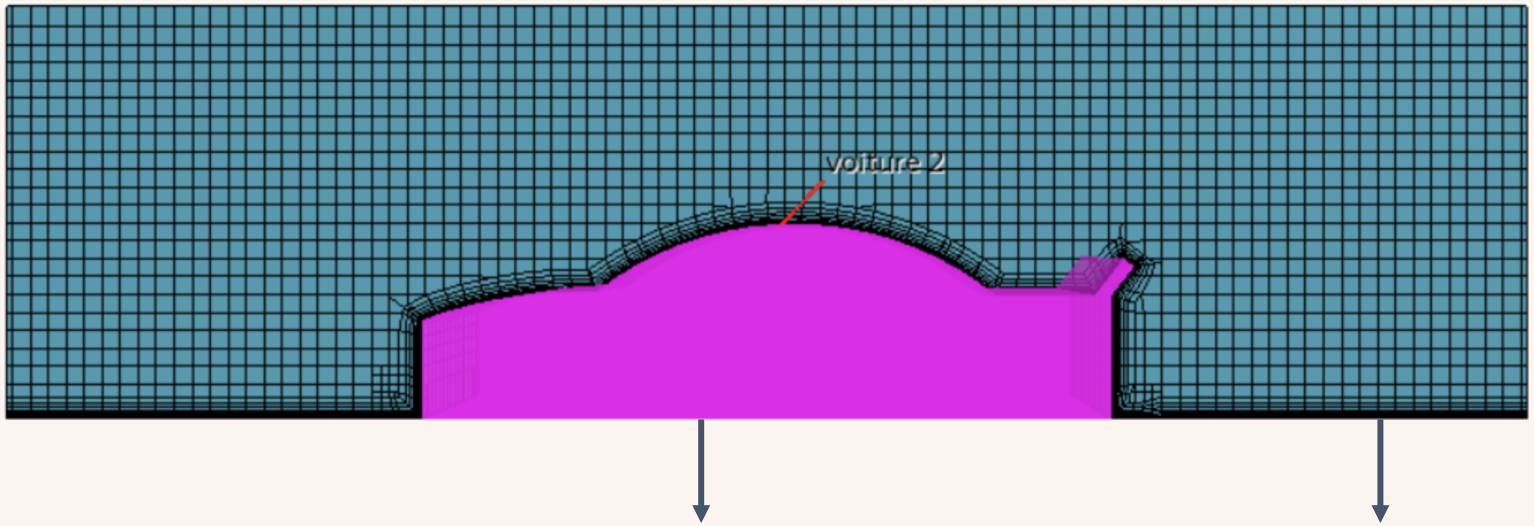
Overset zone

IV. DYNAMIC MODELLING IN 2D/3D
1. Model geometry / mesh / physics

Overset domain

- Base size: 0.13 m
- Number of prism layers: 8
- Prism layer stretching: 1.3
- Prism layer thickness: 0.16 m

- Dimensions: 11 m x 3 m x 1.95 m

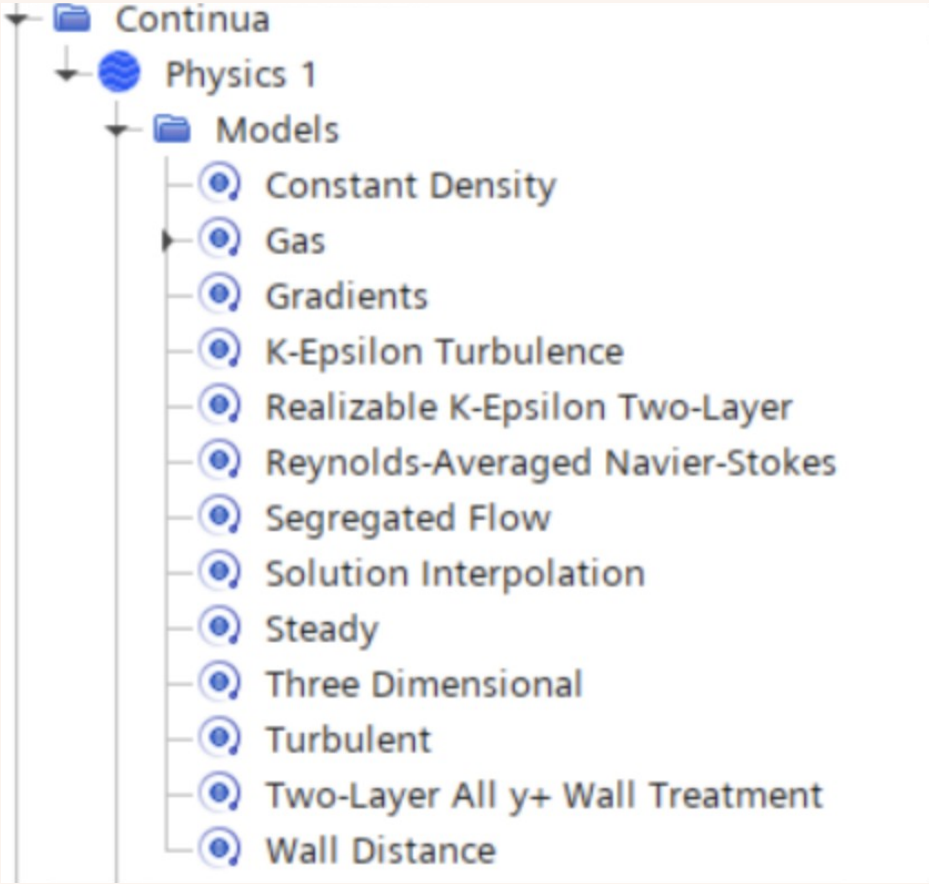


Following car

Overset mesh

IV. DYNAMIC MODELLING IN 2D/3D

1. Model geometry / mesh / physics

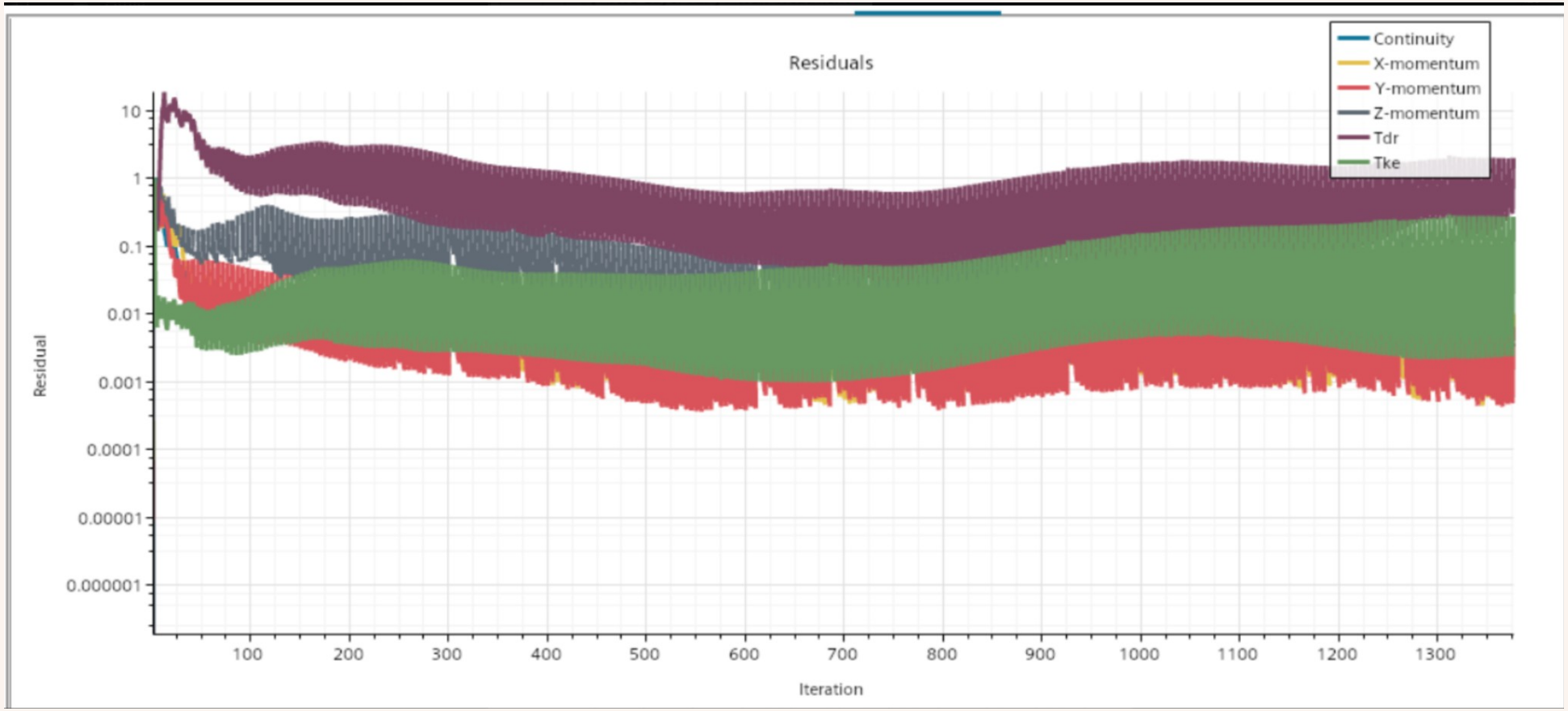


- Constant-density gas (air)
- 3D / 2D unsteady flow
- Overset interface with strict conservation of fluxes/quantities ("Overset Conservation")
- Turbulence modelled with the Realizable two-layer k-epsilon model and "all y+ wall treatment"
- RANS (Reynolds-Averaged Navier-Stokes) solution
- Turbulent regime
- Advanced gradient handling and solution interpolation for robustness
- Segregated equations for numerical stability

IV. DYNAMIC MODELLING IN 2D/3D

2. 3D simulation - qualitative results

- Residual convergence - dynamic 3D model



Residuals at the limit:
Between 0.0002 and 2

Possible causes:
- ➢Heavy 3D mesh
- ➢High velocity
- ➢High turbulence

IV. DYNAMIC MODELLING IN 2D/3D

2. 2D simulation - qualitative results

- Residual convergence - dynamic 2D model

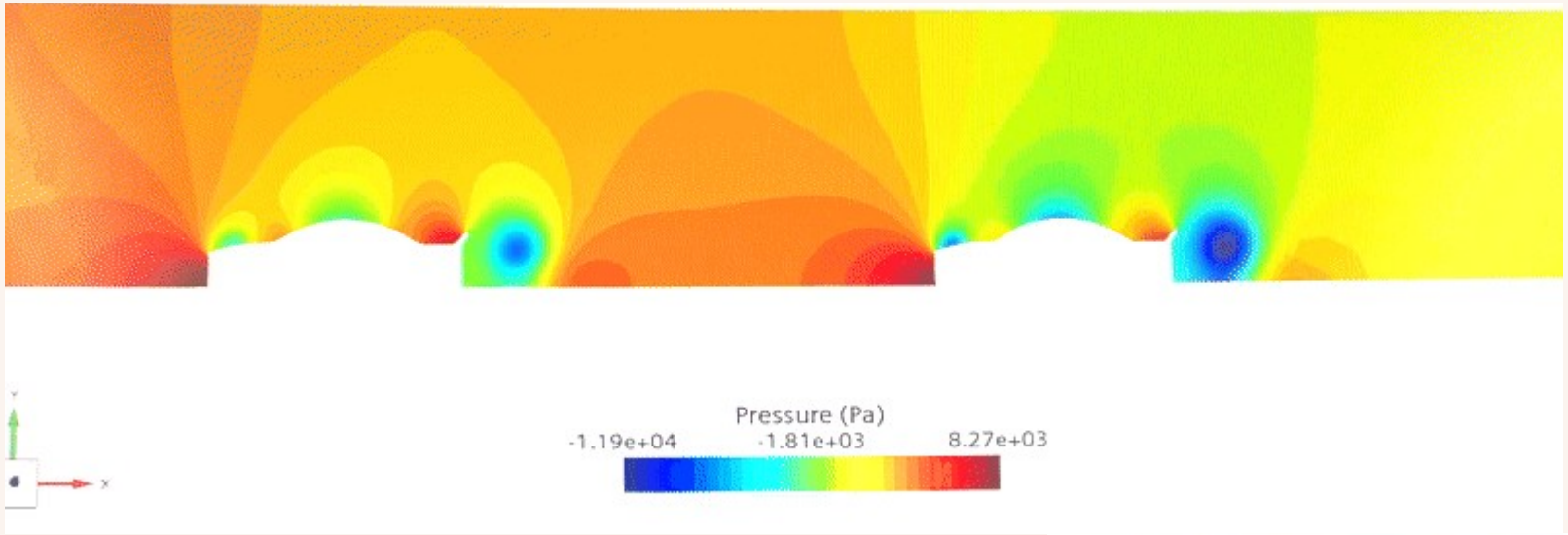


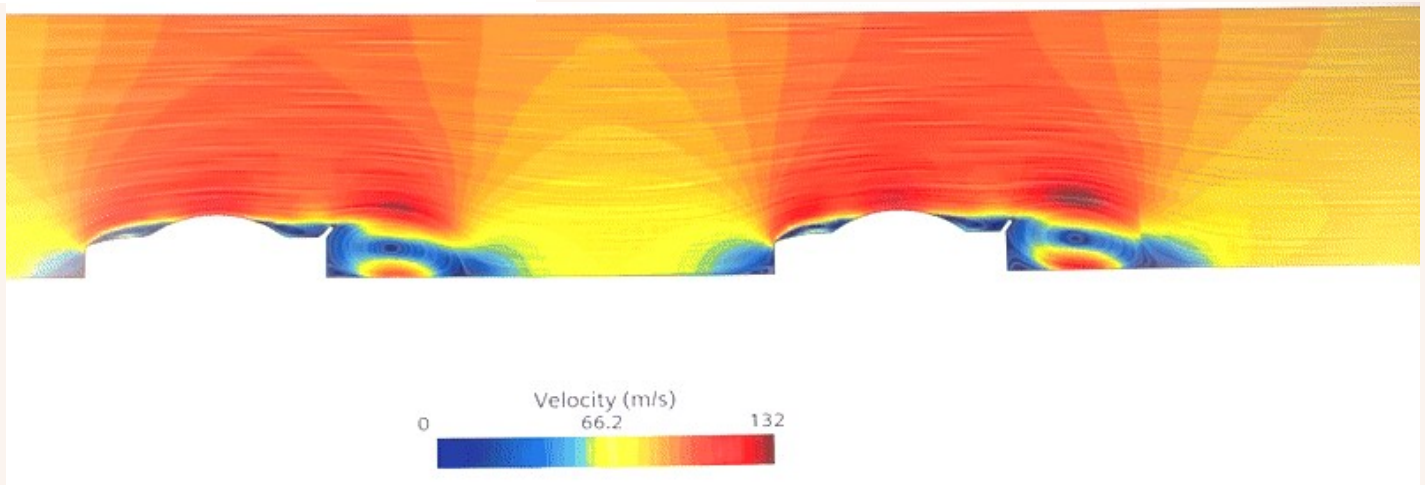
Better residuals:
Between 0.0001 and 0.1

Possible causes:
➢Switch to 2D

IV. DYNAMIC MODELLING IN 2D/3D

2. 2D simulation - qualitative results

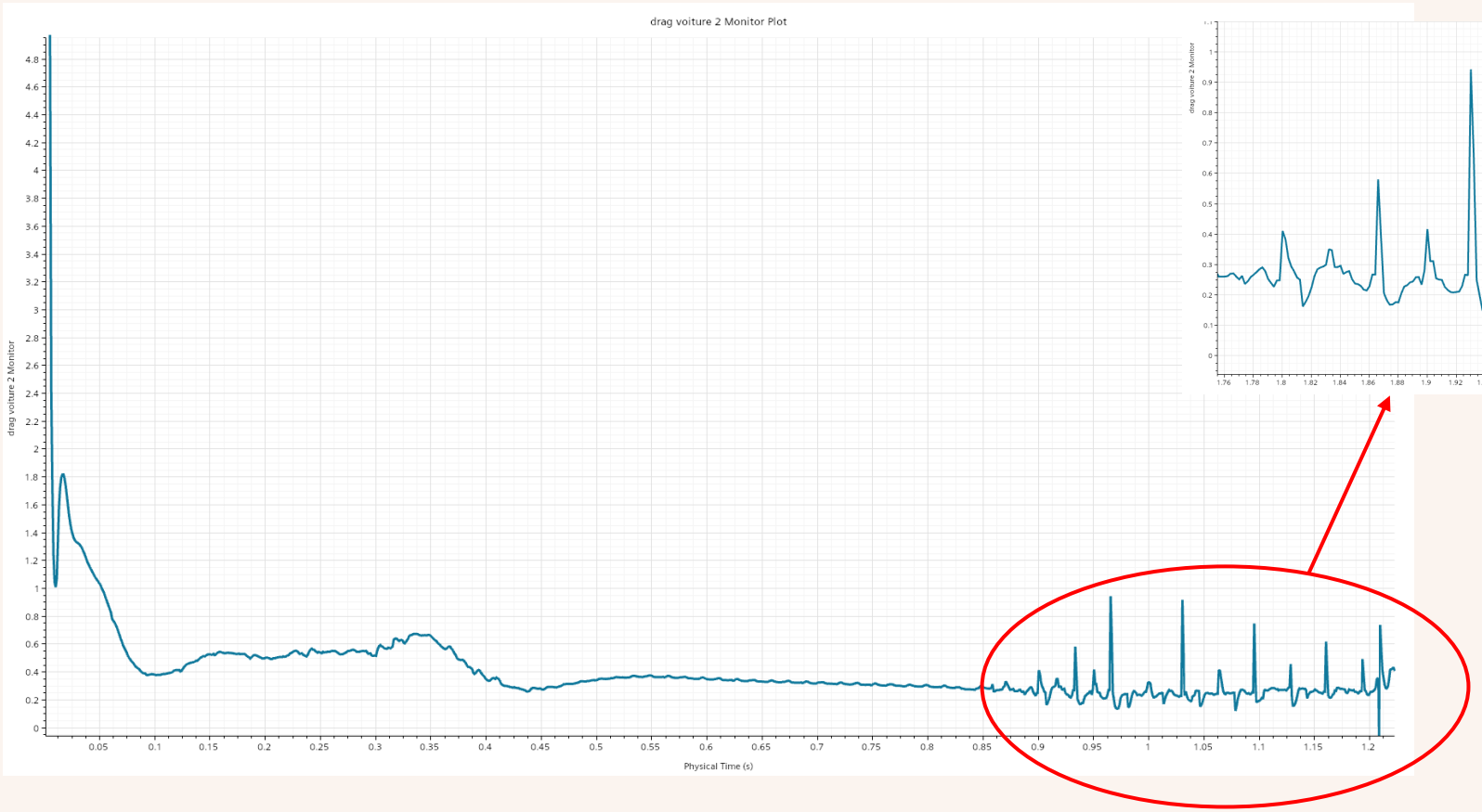


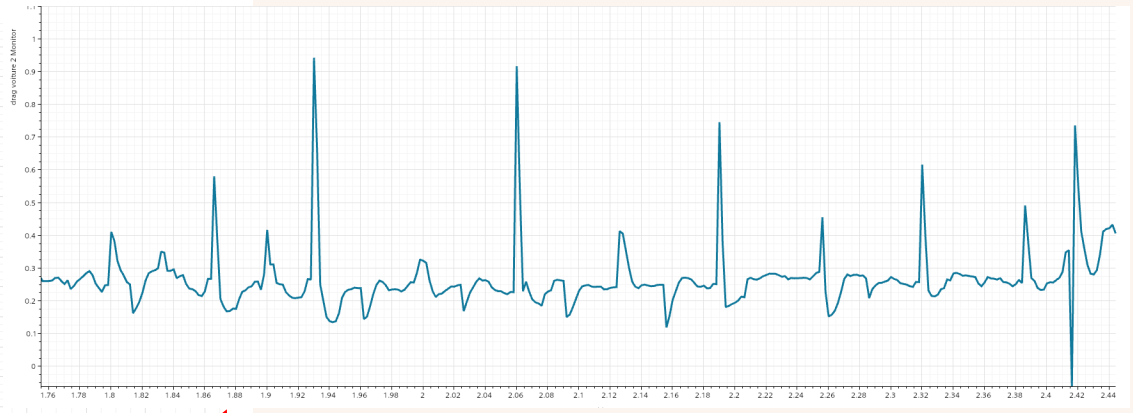


IV. DYNAMIC MODELLING IN 2D/3D

2. 3D simulation - qualitative results

- Static 2D/3D comparison



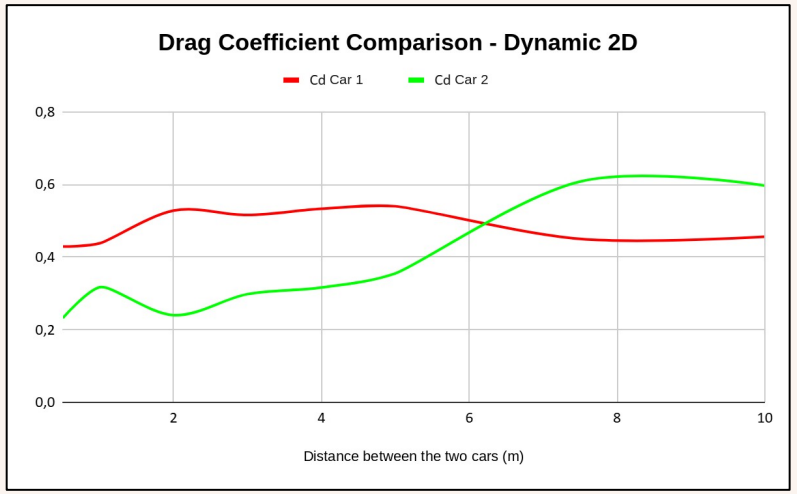


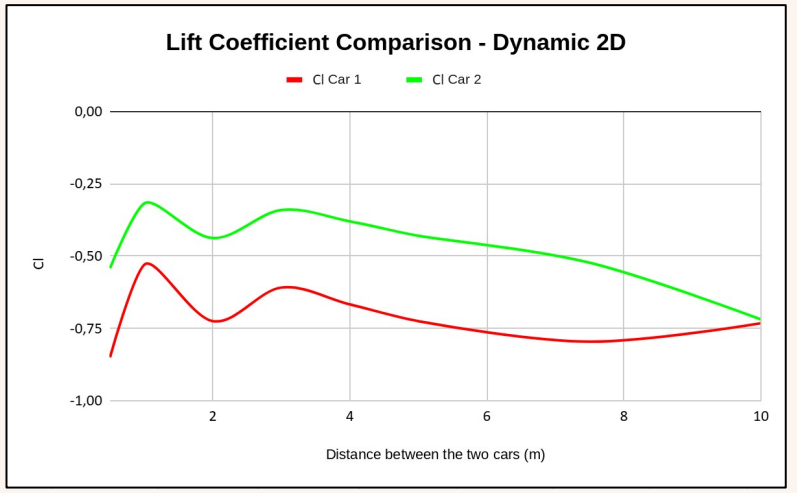
Increase in numerical uncertainty

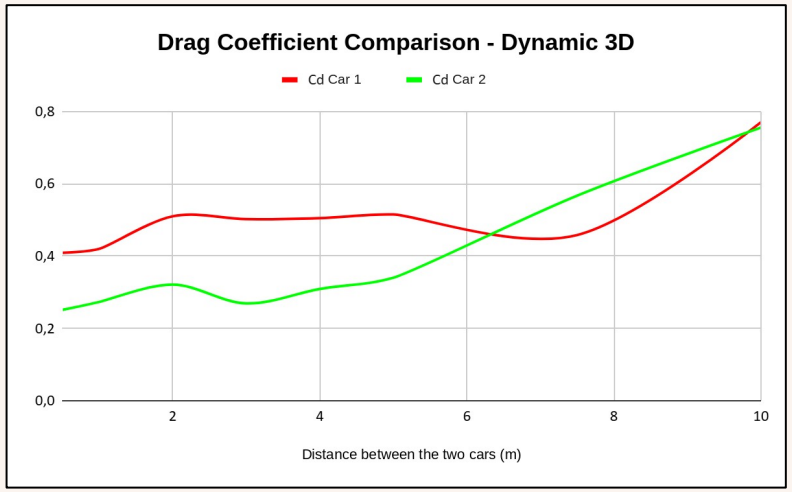
- **Mesh issue**
- **Auto-save**
- **Influence of the transition zone**

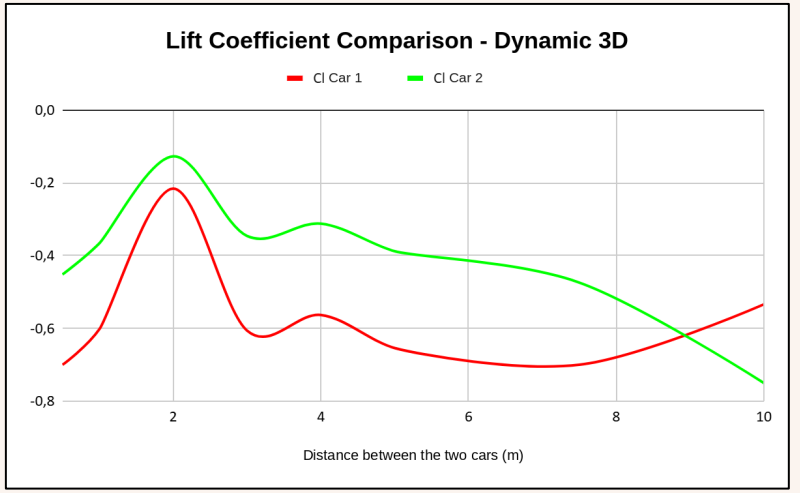
IV. DYNAMIC MODELLING IN 2D/3D

2. 2D simulation - quantitative results



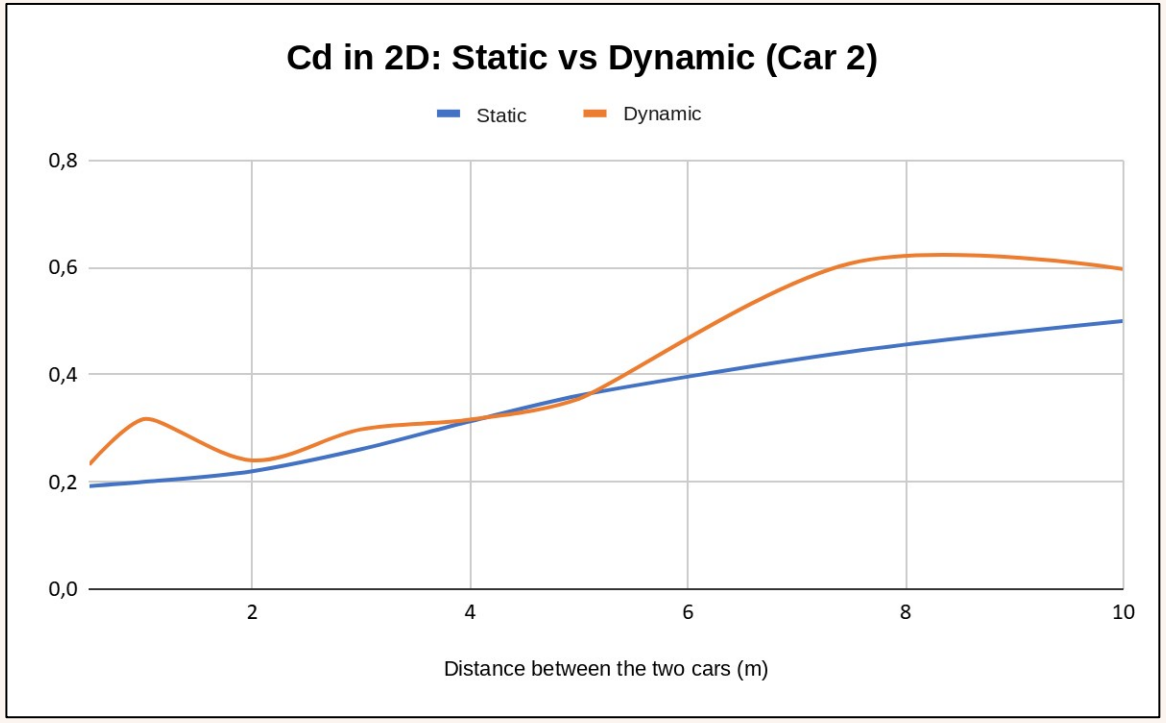


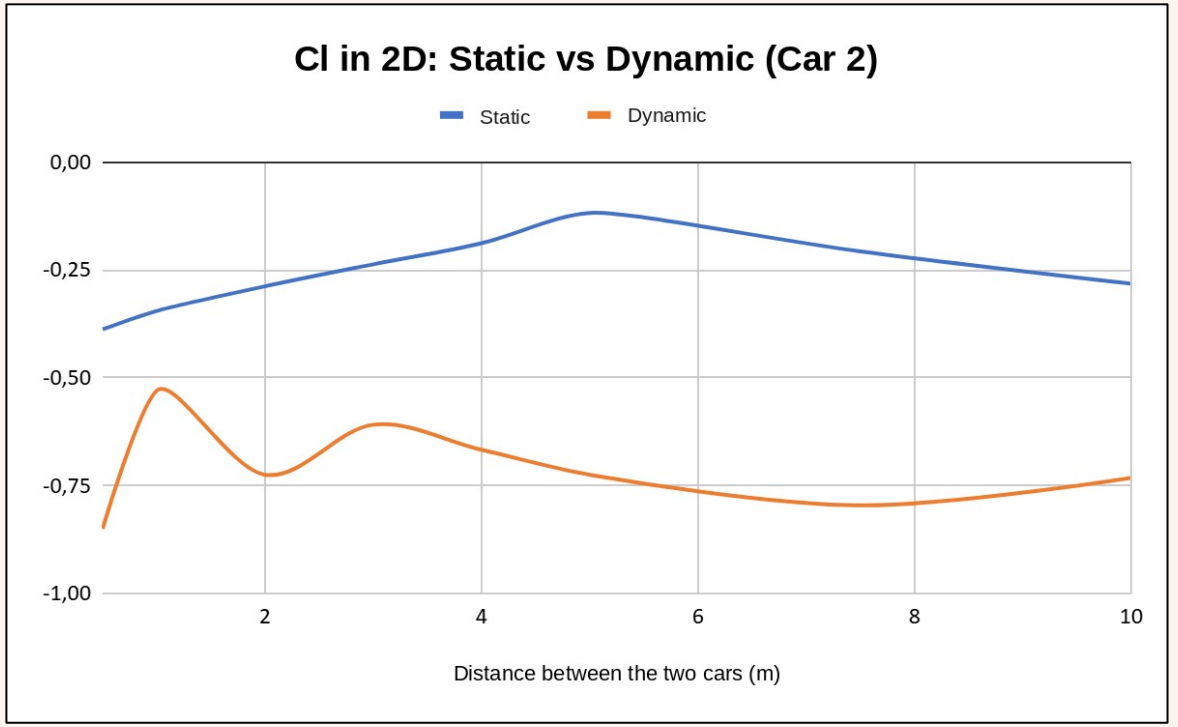




V. COMPARISON OF RESULTS

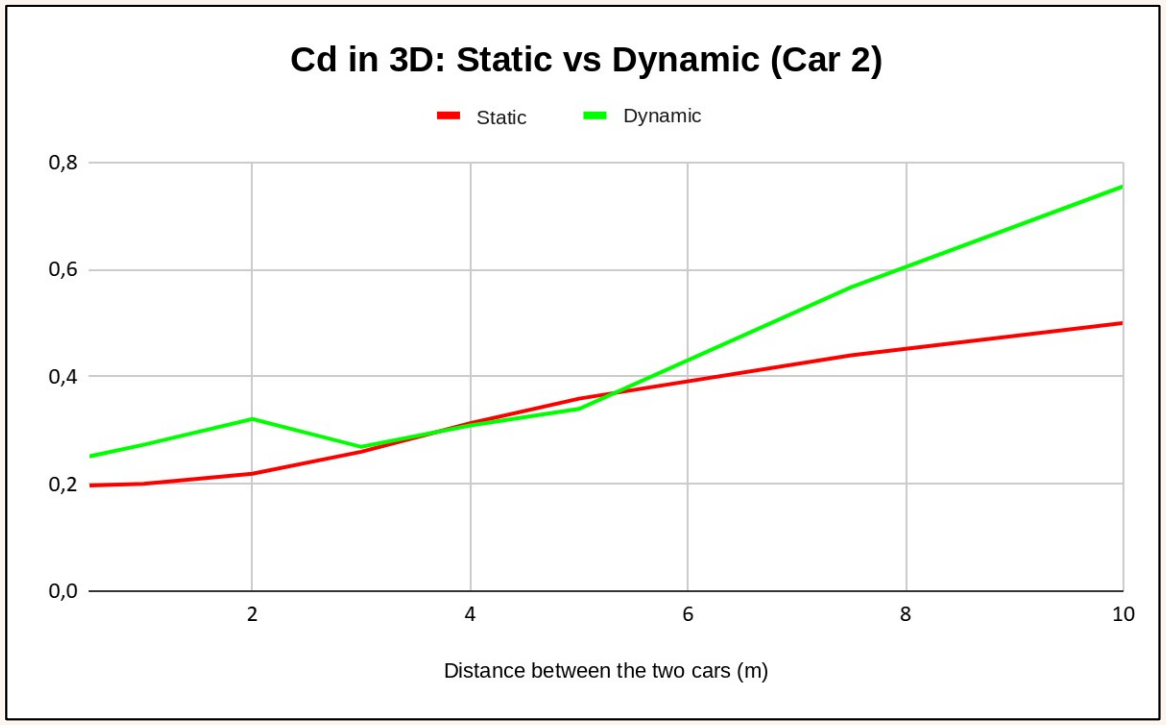
1. Static vs dynamic

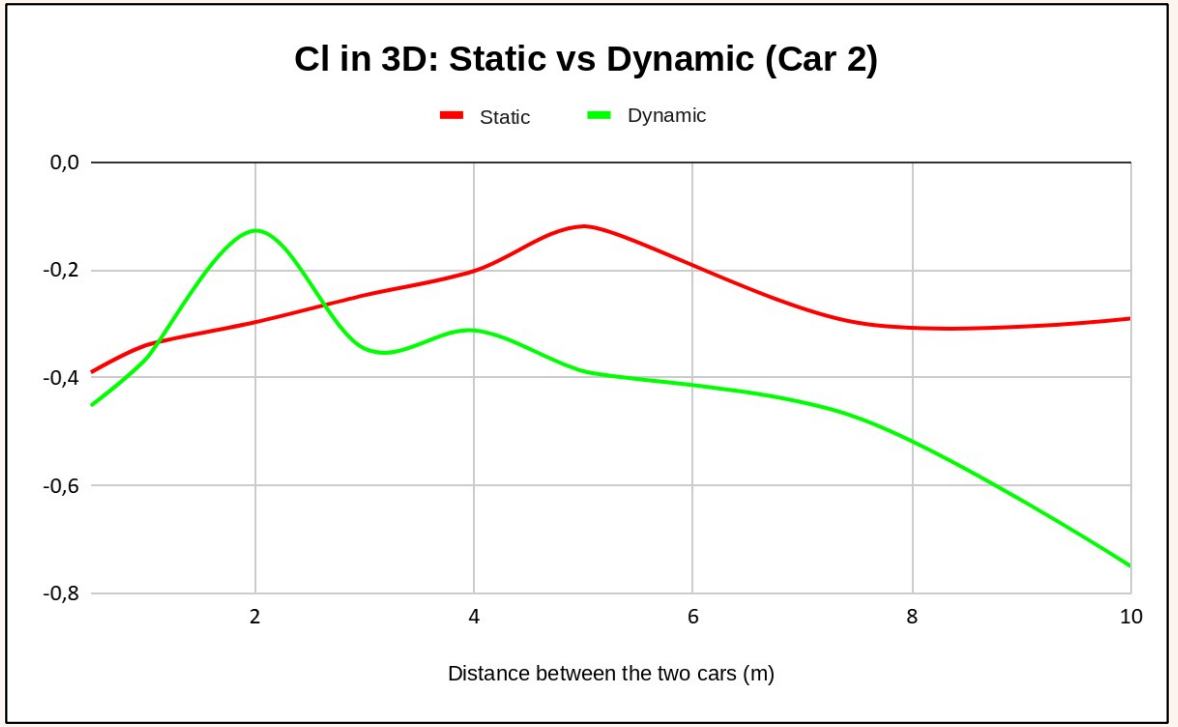




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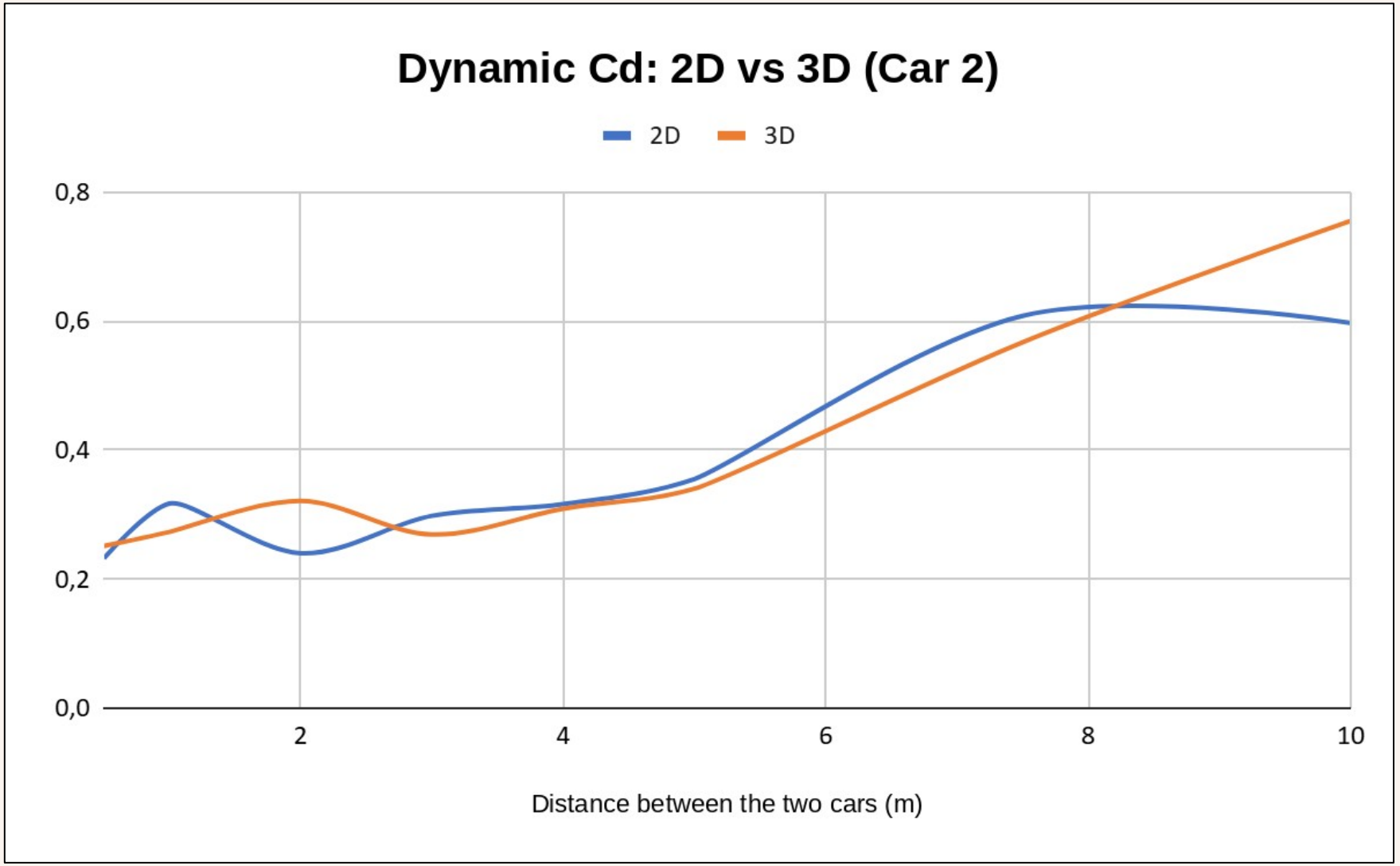
1. Static vs dynamic





V. COMPARISON OF RESULTS

1. 2D vs 3D



VI. CONCLUSION AND OUTLOOK

→ Successful implementation: overset working, Cd/Cl consistent with real NASCAR values

→ Validated contribution: from static to translating dynamic simulation

→ Improvements: 3D mesh + k-ω SST + better geometry

→ Outlook: stall, full overtake, multi-car packs